

Customer Traffic Forecasting for Restaurant Network

Portfolio case study summary - anonymized from internal project documents

T+1 Next-month daily guest-count forecast	XGBoost Core gradient-boosted tree model	Hybrid Global + Local forecasting streams	29.7% Monthly WAPE relative improvement
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Project Overview

This project delivered a production-oriented forecasting pipeline for a large restaurant network. The goal was to predict daily guest traffic for each active restaurant one month ahead, supporting staffing, demand planning, and operating decisions.

The solution improved the previous forecasting flow by strengthening data cleaning, adding restaurant behavior segmentation, expanding feature engineering, and using separate model streams for stable versus high-volatility restaurants.

Problem

- Restaurant traffic varies by weekday, holidays, lunar-calendar effects, seasonality, store maturity, and local volatility.
- A single forecasting approach had difficulty handling both stable restaurants and newer/high-variance restaurants.
- The business needed restaurant-level daily predictions that could also be aggregated to weekly and monthly views.
- Data quality issues such as missing traffic, abnormal guest-count values, and limited history for new restaurants reduced forecast reliability.

Solution

The improved workflow combined robust preprocessing, feature engineering, segmentation, and model optimization into a repeatable monthly forecasting pipeline.

Layer	What was added or improved
Data quality	Invalid-value removal, missing-value handling, and explicit anomaly processing before modeling.
Feature engineering	Time, weekend, holiday, pre-holiday, holiday-weight, lag, rolling-history, restaurant-cluster, and lunar-calendar features.
Segmentation	Restaurant complexity clustering based on weekly seasonality and volatility, enabling different treatment for stable and volatile patterns.
Modeling	Gradient-boosted tree model using XGBoost with rolling monthly training: train through T-1, predict month T.
Forecast design	Dual streams: Global model for lower-complexity restaurants and Local model for higher-complexity restaurants, merged into a final hybrid output.
Pipeline reliability	Cleaner execution flow with monthly run steps for preprocessing, clustering, global prediction, local prediction, and final aggregation.

Model and Feature Details

Component	Portfolio-safe detail
Target	Daily guest count per active restaurant for the next forecast month.
Algorithm	XGBoost / gradient-boosted trees for tabular time-series forecasting.
Training logic	Rolling monthly setup: use historical data up to the previous month, then predict the target forecast month.
Validation	Backtesting over documented historical months using WAPE as the main metric and MAPE as a supporting metric.
Key feature groups	Calendar features, holiday impact, lagged traffic, rolling weekly/monthly statistics, restaurant cluster, and lunar-calendar indicators.
Output	Restaurant-date forecast table plus aggregate performance tracking for daily, weekly, and monthly views.

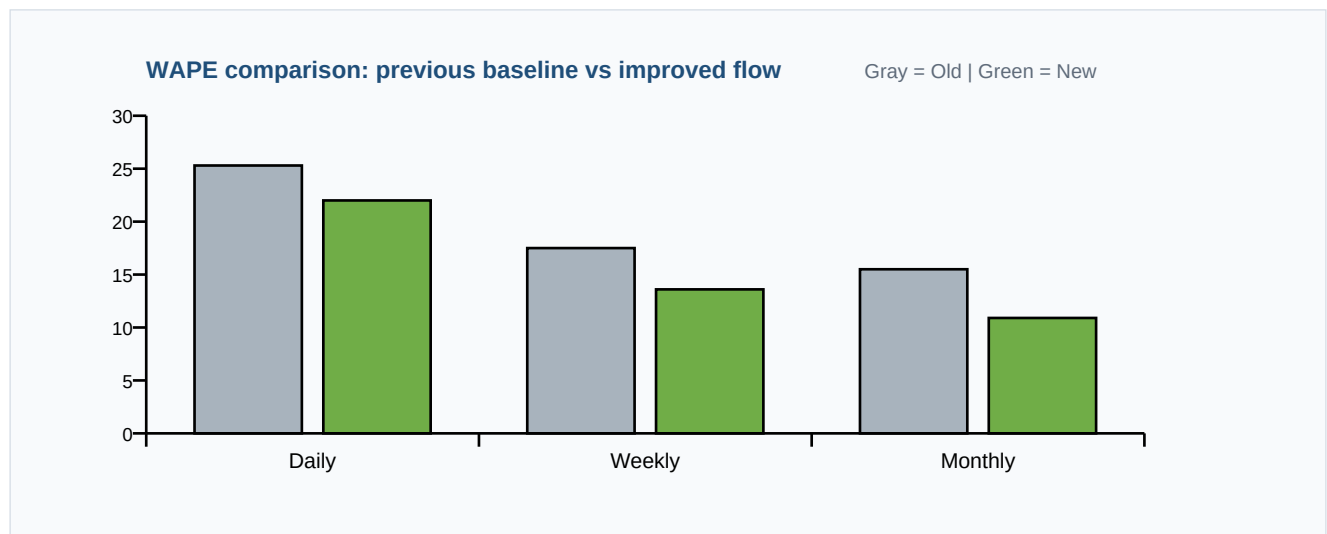
Forecasting Pipeline

1	Preprocess and clean data	Handle invalid, abnormal, and missing traffic records.
2	Engineer features	Create calendar, holiday, lag, rolling-stat, cluster, and lunar features.
3	Segment restaurants	Cluster stores by seasonality and volatility to identify stable versus complex demand patterns.
4	Train and predict	Run Global and Local XGBoost streams for the forecast month.
5	Aggregate hybrid output	Merge, de-duplicate, filter for business rules, and publish final predictions.

Measured Improvement

The new forecasting flow reduced WAPE across daily, weekly, and monthly aggregation levels versus the previous internal baseline. Lower WAPE is better.

Level	Old WAPE	New WAPE	Improvement	Relative improvement
Daily	25.3	22.0	-3.3 pts	13.0%
Weekly	17.5	13.6	-3.9 pts	22.3%
Monthly	15.5	10.9	-4.6 pts	29.7%



Portfolio Impact

- Delivered a more accurate forecasting pipeline, with documented WAPE reductions at daily, weekly, and monthly levels.
- Improved handling of difficult restaurant segments by separating stable and high-volatility patterns into different forecast streams.
- Created a maintainable monthly workflow from preprocessing to final hybrid forecast output.
- Supported business planning with restaurant-level daily forecasts that can be rolled up to operational views.

Confidentiality note: This PDF is a summary from internal documents of a previous company. Private information has been removed or generalized, including client/vendor names, infrastructure paths, account identifiers, and internal environment details.

Suggested portfolio framing: Present this as an anonymized forecasting case study. Do not attach source documents or operational runbooks unless you have explicit permission.